

$$\frac{\cos x}{x^2}. \quad (2 \text{ marks})$$

6. Calculus, 2ADV C2 2006 HSC 2aii

Differentiate $\frac{\sin x}{x+1}$ with respect to x . (2 marks)

7. Calculus, 2ADV C2 2004 HSC 3aii

Differentiate with respect to x :
 $(1 + \sin x)^5$. (2 marks)

8. Calculus, 2ADV C2 2006 HSC 2ai

Differentiate $x \tan x$ with respect to x . (2 marks)

9. Calculus, 2ADV C2 2007 HSC 2aii

Differentiate with respect to x :
 $(1 + \tan x)^{10}$. (2 marks)

10. Calculus, 2ADV C2 2009 HSC 2ai

Differentiate with respect to x :
 $x \sin x$ (2 marks)

11. Calculus, 2ADV C2 2010 HSC 1e

Differentiate $x^2 \tan x$ with respect to x . (2 marks)

12. Calculus, 2ADV C2 2010 HSC 2a

Differentiate $\frac{\cos x}{x}$ with respect to x . (2 marks)

13. Calculus, 2ADV C2 2011 HSC 4a

Differentiate $\frac{x}{\sin x}$ with respect to x . (2 marks)

14. Calculus, 2ADV C2 2013 HSC 11c

Differentiate $(\sin x - 1)^8$. (2 marks)

15. Calculus, 2ADV C2 2017 HSC 11c

Differentiate $\frac{\sin x}{x}$. (2 marks)

16. Calculus, 2ADV C2 2008 HSC 2aiii

Differentiate with respect to x :
 $\frac{\sin x}{x+4}$. (2 marks)

17. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :
 $e^{\tan(2x)}$ (2 marks)

18. Calculus, 2ADV C2 2019 MET1 1b

Let $f(x) = x^2 \cos(3x)$.
Find $f'\left(\frac{\pi}{3}\right)$. (2 marks)

19. Calculus, 2ADV C2 2022 HSC 25

Let $f(x) = \sin(2x)$.
Find the value of x , for $0 < x < \pi$, for which $f'(x) = -\sqrt{3}$ AND $f''(x) = 2$. (3 marks)

20. Calculus, 2ADV C2 EQ-Bank 10

The function $f(\theta) = \sin^3(2\theta)$.
If $f'(\theta) = 6\cos(2\theta) - 6\cos^n(2\theta)$, find the value of n . (2 marks)

Worked Solutions

1. Calculus, 2ADV C2 2016 HSC 5 MC

$$\begin{aligned}y &= \ln(\cos x) \\ \frac{dy}{dx} &= \frac{-\sin x}{\cos x} \\ &= -\tan x \\ &\Rightarrow B\end{aligned}$$

2. Calculus, 2ADV C4 2018 HSC 5 MC

$$\begin{aligned}y &= \sin(\ln x) \\ \frac{dy}{dx} &= \cos(\ln x) \times \frac{d}{dx}(\ln x) \\ &= \cos(\ln x) \times \frac{1}{x} \\ &= \frac{\cos(\ln x)}{x} \\ &\Rightarrow D\end{aligned}$$

3. Calculus, 2ADV C2 2013 HSC 4 MC

$$\begin{aligned}y &= \frac{x}{\cos x} \\ \text{Let } u &= x & u' &= 1 \\ v &= \cos x & v' &= -\sin x \\ \therefore \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{\cos x \cdot 1 - x(-\sin x)}{(\cos x)^2} \\ &= \frac{\cos x + x\sin x}{\cos^2 x} \\ &\Rightarrow A\end{aligned}$$

4. Calculus, 2ADV C2 2019 HSC 11b

Using the product rule:

$$\frac{d}{dx}(x^2 \sin x) = x^2 \cdot \cos x + 2x \sin x$$

Worked Solutions

5. Calculus, 2ADV C2 2012 HSC 12ai

$$\begin{aligned}y &= \frac{\cos x}{x^2} \\ u &= \cos x & u' &= -\sin x \\ v &= x^2 & v' &= 2x\end{aligned}$$

Using the quotient rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - v'u}{v^2} \\ &= \frac{-\sin x \cdot x^2 - 2x \cdot \cos x}{x^4} \\ &= \frac{-x\sin x - 2\cos x}{x^3}\end{aligned}$$

6. Calculus, 2ADV C2 2006 HSC 2ai

$$\begin{aligned}y &= \frac{\sin x}{x+1} \\ \text{Using } \frac{d}{dx}\left(\frac{u}{v}\right) &= \frac{u'v - uv'}{v^2} \\ u &= \sin x & v &= x+1 \\ u' &= \cos x & v' &= 1 \\ \therefore \frac{dy}{dx} &= \frac{\cos x(x+1) - \sin x}{(x+1)^2}\end{aligned}$$

7. Calculus, 2ADV C2 2004 HSC 3ai

$$\begin{aligned}y &= (1 + \sin x)^5 \\ \frac{dy}{dx} &= 5(1 + \sin x)^4 \times \frac{d}{dx}(\sin x) \\ &= 5\cos x(1 + \sin x)^4\end{aligned}$$

8. Calculus, 2ADV C2 2006 HSC 2ai

- i. $y = x \tan x$

Using product rule

$$\begin{aligned}\frac{d}{dx}(uv) &= u'v + uv' \\ \therefore \frac{dy}{dx} &= \tan x + x \times \sec^2 x \\ &= x\sec^2 x + \tan x\end{aligned}$$

9. Calculus, 2ADV C2 2007 HSC 2ai

$$y = (1 + \tan x)^{10}$$

$$\begin{aligned}\frac{dy}{dx} &= 10(1 + \tan x)^9 \times \frac{d}{dx}(\tan x) \\ &= 10\sec^2 x (1 + \tan x)^9\end{aligned}$$

10. Calculus, 2ADV C2 2009 HSC 2ai

$$y = x \sin x$$

$$\begin{aligned}\frac{dy}{dx} &= x \cos x + \sin x \times 1 \\ &= x \cos x + \sin x\end{aligned}$$

11. Calculus, 2ADV C2 2010 HSC 1e

$$y = x^2 \tan x$$

Using product rule:

$$\begin{aligned}\frac{d}{dx}(uv) &= u'v + uv' \\ \frac{dy}{dx} &= 2x \tan x + x^2 \sec^2 x\end{aligned}$$

COMMENT: There is no necessity to take out the common factor x in this case.

12. Calculus, 2ADV C2 2010 HSC 2a

$$y = \frac{\cos x}{x}$$

$$\begin{aligned}\text{Let } u &= \cos x & u' &= -\sin x \\ v &= x & v' &= 1\end{aligned}$$

Using quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{-\sin x \cdot x - \cos x \cdot 1}{x^2} \\ &= \frac{-x \sin x - \cos x}{x^2}\end{aligned}$$

MARKER'S COMMENT: A significant number of students did not know the quotient rule and many found assistance by writing out u , u' , v , v' before going into calculations.

13. Calculus, 2ADV C2 2011 HSC 4a

$$y = \frac{x}{\sin x}$$

$$\begin{aligned}u &= x & u' &= 1 \\ v &= \sin x & v' &= \cos x\end{aligned}$$

$$\text{Using } \frac{d}{dx}(uv) = \frac{u'v - uv'}{v^2},$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1 \cdot \sin x - x \cdot \cos x}{(\sin x)^2} \\ &= \frac{\sin x - x \cos x}{\sin^2 x}\end{aligned}$$

14. Calculus, 2ADV C2 2013 HSC 11c

$$y = (\sin x - 1)^8$$

$$\begin{aligned}\frac{dy}{dx} &= 8(\sin x - 1)^7 \times \frac{d}{dx}(\sin x - 1) \\ &= 8(\sin x - 1)^7 \times \cos x \\ &= 8 \cos x (\sin x - 1)^7\end{aligned}$$

15. Calculus, 2ADV C2 2017 HSC 11c

$$y = \frac{\sin x}{x}$$

$$\begin{aligned}\text{Let } u &= \sin x & u' &= \cos x \\ v &= x & v' &= 1\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{x \cos x - \sin x}{x^2}\end{aligned}$$

16. Calculus, 2ADV C2 2008 HSC 2aiii

$$y = \frac{\sin x}{x + 4}$$

$$\begin{aligned}u &= \sin x & u' &= \cos x \\ v &= x + 4 & v' &= 1\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{\cos x (x + 4) - \sin x}{(x + 4)^2}\end{aligned}$$

17. Calculus, 2ADV C2 EQ-Bank 2

$$\begin{aligned}y &= e^{\tan(2x)} \\ \frac{dy}{dx} &= \frac{d}{dx} \tan(2x) \times e^{\tan(2x)} \\ &= 2\sec^2(2x) \cdot e^{\tan(2x)}\end{aligned}$$

18. Calculus, 2ADV C2 2019 MET1 1b

$$\begin{aligned}f(x) &= x^2\cos3x \\ f'(x) &= x^2 \cdot 3(-\sin3x) + 2x\cos3x \\ f'\left(\frac{\pi}{3}\right) &= \left(\frac{\pi}{3}\right)^2 \cdot 3(-\sin\pi) + 2\left(\frac{\pi}{3}\right)\cos\pi \\ &= -\frac{2\pi}{3}\end{aligned}$$

19. Calculus, 2ADV C2 2022 HSC 25

$$\begin{aligned}f'(x) &= 2\cos(2x) \\ 2\cos(2x) &= -\sqrt{3} \\ \cos(2x) &= -\frac{\sqrt{3}}{2} \\ 2x &= \pi - \frac{\pi}{6}, \quad \pi + \frac{\pi}{6} \\ &= \frac{5\pi}{6}, \quad \frac{7\pi}{6} \\ x &= \frac{5\pi}{12}, \quad \frac{7\pi}{12}\end{aligned}$$

$$\begin{aligned}f''(x) &= -4\sin(2x) \\ -4\sin(2x) &= 2 \\ \sin(2x) &= -\frac{1}{2} \\ 2x &= \pi + \frac{\pi}{6}, \quad 2\pi - \frac{\pi}{6} \\ &= \frac{7\pi}{6}, \quad \frac{11\pi}{6} \\ x &= \frac{7\pi}{12}, \quad \frac{22\pi}{12}\end{aligned}$$

$$\therefore x = \frac{7\pi}{12} \text{ (satisfies both equations)}$$

20. Calculus, 2ADV C2 EQ-Bank 10

$$\begin{aligned}f(\theta) &= \sin^3(2\theta) \\ &= (\sin(2\theta))^3\end{aligned}$$

$$\begin{aligned}f'(\theta) &= 3 \times 2\cos(2\theta) \times \sin^2(2\theta) \\ &= 6\cos(2\theta)(1-\cos^2(2\theta)) \\ &= 6\cos(2\theta)-6\cos^3(2\theta)\end{aligned}$$

$$\therefore n = 3$$